DCCA – 501

V Semester B.C.A. Degree Examination, February/March 2024
(NEP) (Freshers)
COMPUTER APPLICATIONS
Design and Analysis of Algorithms

Time : 2½ Hours Max. Marks : 60

Instruction : *Answer **all** the Sections.*

SECTION – A

I. Answer **any six** questions. **Each** question carries **2** marks. **(6×2=12)**

1) Define order of growth.
2) Write an algorithm to compute gcd of two numbers.
3) What do you mean by recursive algorithm ?
4) What are the various factors that affect the execution time ?
5) Write any two advantages of selection sort.
6) What is brute force approach ?
7) What is the concept of decrease and conquer methodology ?
8) Write a short note on greedy algorithm.
9) What is NP-class ?

SECTION – B

II. Answer **any four** questions. **Each** question carries **6** marks. **(4×6=24)**

10) Explain the fundamentals of algorithmic problem solving.
11) Write a general plan for analyzing non-recursive algorithm.
12) Explain the TSP with a suitable example.
13) a) What is knapsack problem ? **2**

    b) Write any two advantages and disadvantages of divide and conquer technique. **4**
14) Explain merge sort algorithm with an example.
15) Write a program that implements Prim's algorithm to generate minimum cost spanning tree.

P.T.O.

SECTION – C

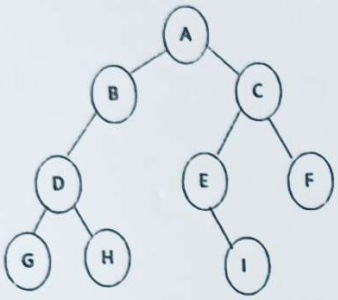
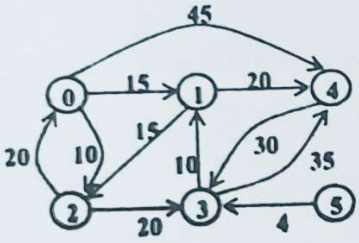
III. Answer **any three** questions. **Each** question carries **8** marks. **(3×8=24)**

16) Define algorithm. What are the criteria that an algorithm must satisfy ?

17) Explain asymptotic notations.

18) Compare and contrast BFS and DFS.

19) Define tree. Traverse the following tree in pre-order, post-order and in-order.



20) Obtain the shortest distance and shortest path from node 5 to node 1 in the following graph :